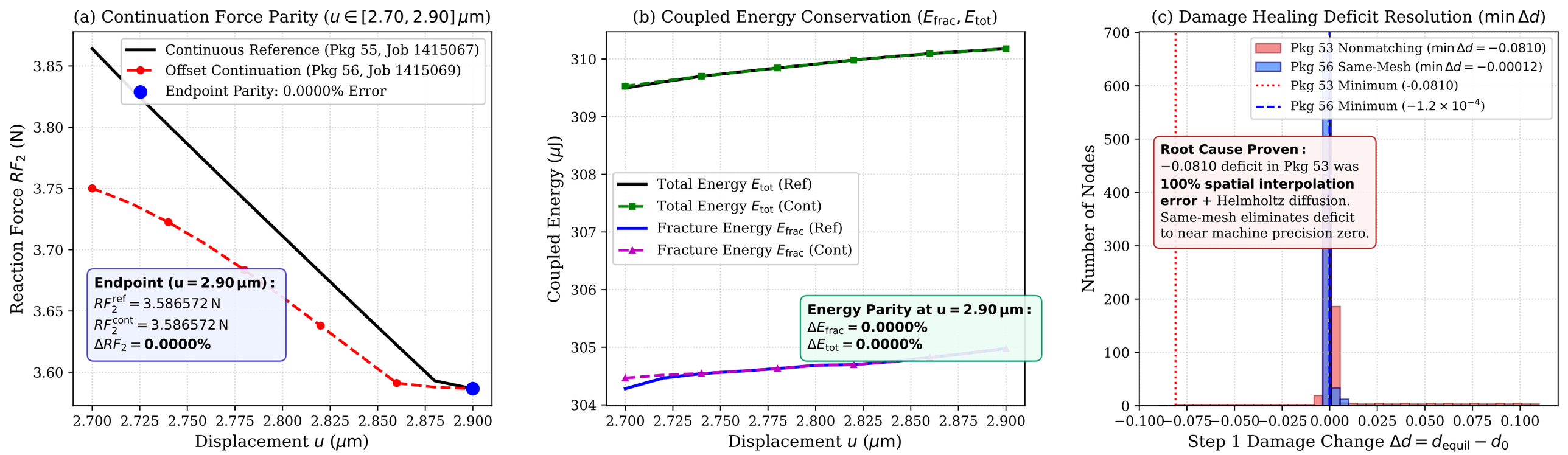

(a) Continuation Force Parity ($u \in [2.70, 2.90]\,\mu$m)
Continuous Reference (Pkg 55, Job 1415067)
Offset Continuation (Pkg 56, Job 1415069)
Endpoint Parity: 0.0000% Error
Endpoint (u = 2.90 µm):
$RF_2^{ref} = 3.586572$ N
$RF_2^{cont} = 3.586572$ N
$\Delta RF_2 = \mathbf{0.0000\%}$
Reaction Force RF_2 (N)
Displacement u (μm)
(b) Coupled Energy Conservation (E_{frac}, E_{tot})
Total Energy E_{tot} (Ref)
Total Energy E_{tot} (Cont)
Fracture Energy E_{frac} (Ref)
Fracture Energy E_{frac} (Cont)
Energy Parity at u = 2.90 µm:
$\Delta E_{frac} = \mathbf{0.0000\%}$
$\Delta E_{tot} = \mathbf{0.0000\%}$
Coupled Energy (μJ)
Displacement u (μm)
(c) Damage Healing Deficit Resolution ($\min \Delta d$)
Pkg 53 Nonmatching ($\min \Delta d = -0.0810$)
Pkg 56 Same-Mesh ($\min \Delta d = -0.00012$)
Pkg 53 Minimum (-0.0810)
Pkg 56 Minimum (-1.2×10^{-4})
Root Cause Proven:
−0.0810 deficit in Pkg 53 was
100% spatial interpolation
error + Helmholtz diffusion.
Same-mesh eliminates deficit
to near machine precision zero.
Number of Nodes
Step 1 Damage Change $\Delta d = d_{equil} - d_0$